



Bahria University
Discovering Knowledge
Karachi Campus

CSL-220: Database Management Systems Lab

Lab # 04 Multiple Row Functions

Instructor: Engr. Hareem Fatima

SQL Multiple-Row Functions (Aggregate Functions)

- Operate on multiple rows as a group.
- Returns a single result for the entire group.
- Used to perform calculations on groups of rows, often in conjunction with the GROUP BY and HAVING clauses of the SELECT statement.

Useful aggregate functions:

- **AVG()** - Returns the average value
- **COUNT()** - Returns the number of rows
- **MAX()** - Returns the largest value
- **MIN()** - Returns the smallest value
- **SUM()** - Returns the sum

Examples

	EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO	GENDER
1	7369	HAIDER	CLERK	7566	1983-06-13 00:00:00	800.00	NULL	20	M
2	7499	AWAIS	SALESMAN	7698	1983-08-15 00:00:00	1600.00	400.00	30	M
3	7521	SHEHROZ	SALESMAN	7698	1984-03-26 00:00:00	1250.00	300.00	30	M
4	7566	KHALID	MANAGER	7839	1983-10-31 00:00:00	2975.00	NULL	20	M
5	7654	KATERINA	SALESMAN	7698	1983-12-05 00:00:00	1250.00	1400.00	30	F
6	7698	SAFDAR	MANAGER	7839	1984-06-11 00:00:00	2850.00	NULL	30	M
7	7782	SHUJAAT	MANAGER	7839	1984-05-14 00:00:00	2450.00	NULL	10	M
8	7788	FAIZAN	ANALYST	7566	1984-03-05 00:00:00	3000.00	NULL	20	M
9	7839	BASHIR	PRESIDENT	NULL	1984-07-09 00:00:00	5000.00	NULL	NULL	M
10	7844	RUDABA	SALESMAN	7698	1984-06-04 00:00:00	1500.00	500.00	30	F
11	7876	MOHSIN	CLERK	7788	1984-06-04 00:00:00	1100.00	NULL	20	M
12	7900	NELSON	CLERK	7698	1984-07-23 00:00:00	950.00	NULL	30	M
13	7902	LUBNA	ANALYST	7566	1983-12-05 00:00:00	3000.00	NULL	20	F
14	7934	ADNAN	CLERK	7782	1983-11-21 00:00:00	1300.00	NULL	10	M
15	7935	M ALI	NETWORK ADMIN	7839	2000-09-01 00:00:00	4000.00	NULL	50	M

Select count(comm) from emp
where deptno = 30;

(No column name)	
1	4

Select avg(sal) AS AvgSalary, max(sal) AS
MaxSalary, sum(sal) AS TotalSalary from emp where
job like 'SALES%';

	AvgSalary	MaxSalary	TotalSalary
1	1400.000000	1600.00	5600.00

Select count(distinct(deptno)) AS numberofdepts
from emp;

	numberofdepts
1	4

Difference ?

```
SELECT COUNT(*) FROM emp;
```

	(No column name)
1	15

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```
SELECT COUNT(DISTINCT DEPTNO) FROM emp;
```

	(No column name)
1	4

```
SELECT COUNT(ALL DEPTNO) FROM emp;
```

	(No column name)
1	14

SQL GROUP BY

The **SQL GROUP BY** statement is used along with the SQL aggregate functions to organize rows with similar values in specific columns together, It's like putting similar items into categories, then performing calculations or aggregations on each group separately.

Example 1 : find number of employees against each job title in every department.

```
Select deptno, job, count(EMPNO/*)  
as numberofemployees  
from emp  
group by job, deptno;
```

	deptno	job	numberofemployees
1	NULL	PRESIDENT	1
2	10	CLERK	1
3	10	MANAGER	1
4	20	ANALYST	2
5	20	CLERK	2
6	20	MANAGER	1
7	30	CLERK	1
8	30	MANAGER	1
9	30	SALESMAN	4
10	50	NETWORK ADMIN	1

SQL GROUP BY

Example 2: We want to display the total salary being paid to each job title, within each department.

Try yourself 😊

SQL HAVING CLAUSE

The **HAVING** clause in SQL is used in conjunction with the **GROUP BY** clause to filter the results of a query based on aggregate conditions.

While the **WHERE** clause is used to filter individual rows before grouping, the **HAVING** clause is used to filter groups of rows after they have been grouped and aggregated.

Examples:

1. Find Departments with Total Salary Greater Than 10,000

```
SELECT DEPTNO, SUM(SAL) AS TotalSalary
FROM Emp
GROUP BY DEPTNO
HAVING SUM(SAL) > 10000;
```

	DEPTNO	TotalSalary
1	20	10875.00

SQL HAVING CLAUSE

2. Show Jobs with Average Salary Above 2000

```
SELECT JOB, AVG(SAL) AS AverageSalary  
FROM Emp  
GROUP BY JOB  
HAVING AVG(SAL) > 2000;
```

	JOB	AverageSalary
1	ANALYST	3000.000000
2	MANAGER	2758.333333
3	NETWORK ADMIN	4000.000000
4	PRESIDENT	5000.000000

3. Show Jobs with More Than 2 Employees and Average Salary Above 2500.

Try yourself 😊